Topological spaces

topology A set T of subsets of X such that

- $X \in T$
- $\varnothing \in T$
- T is closed under *arbitrary unions* and *finite intersections*

basis A collection $\mathcal{B}$ of subsets of X such that

- $\forall x \in X, \exists B \in \mathcal{B} : x \in B$
- $\forall B_1, B_2 \in \mathcal{B}, x \in B_1 \cap B_2 \implies \exists B_3 \in \mathcal{B} : x \in B_3 \subseteq B_1 \cap B_2$

The set $T_{\mathcal{B}} = \{\bigcup \mathcal{F} \mid \mathcal{F} \subset \mathcal{B}\}$ is the topology generated by $\mathcal{B}$.

fineness T is finer than $S \iff S \subset T$

subspace (Y, T_Y) where $Y \subset X$ and $T_Y = \{Y \cap U \mid U \in T\}$

neighborhood U is a neighborhood of $x \iff$ U is an open set containing x

closure $\overline{Y} := \mathrm{Cl}_X(Y) := \bigcap \{U \mid \complement_X U \in T \wedge Y \subset U\}$ is the smallest closed set containing Y.

interior $\mathrm{Int}(Y) := \bigcup \{U \mid U \in T \wedge U \subset Y\}$ is the largest open set contained in Y

boundary $\partial_X Y := \mathrm{Cl}(Y) \setminus \mathrm{Int}(Y)$

boundary point point x in subspace Y is a *boundary point* of Y $\iff x \in \partial_X Y$

limit point point x in subspace Y is a *limit point* of Y $\iff \forall$open neighborhood U of $x, U \cap Y \neq \varnothing$. Y' denotes the set of limit points of Y.

isolated point point x in subspace Y is an *isolated point* of Y $\iff \{x\}$ is open in Y

limit of sequence x for a sequence $(x_n)_{n\in\mathbb{N}}$ of points in X $\iff \forall U \in T, x \in U \implies \exists N \in \mathbb{N} : \forall n \geq N, x_n \in U$. Equivalently though informally, there comes a point where the terms of the sequence remain within a neighborhood of x.

convergence existence of the limit of a sequence

pointwise convergence for a sequence of functions $(f_i)_{i\in\mathbb{N}}$, if $(f_i(x))_{i\in\mathbb{N}}$ converges for all x in the domain of f_i.

uniform convergence $\forall \varepsilon > 0, \exists N \in \mathbb{N} : \forall n > N, \forall x \in X, d(f_n(x), f(x)) < \varepsilon$. converges to f

Properties

connected if $\nexists$ open sets $U, V : U \cup V = X \wedge U \cap V = \varnothing$, or equivalently X and $\varnothing$ are the only clopen sets

path-connected if $\forall p, q \in X, \exists$ continuous $(\phi : [0,1] \to X) : \phi(0) = p \wedge \phi(1) = q$. Implies but is not implied by connectedness.

connected components the equivalence class with respect to the relation $x \sim y \iff x, y \in$ connected subspace Y of X

path-connected components the equivalence class with respect to the relation $x \sim y \iff \exists$path ϕ between x & y

compact given that a *cover* of $Y \subseteq X$ is a collection $\mathcal{C}$ of subsets of X such that $\bigcup \mathcal{C} \supseteq Y$, and an *open cover* is a cover whose elements are open, Y is *compact* if every open cover of Y includes a finite open cover of Y.

limit point compact if every infinite subset of X has a limit point. Implied by but does not imply compactness.

Separation axioms

For a topological space X in which *singletons are closed*, X is said to be

regular if every pair of disjoint point and closed set is separated by neighborhoods

- $\implies$ subspaces of X are also regular
- $\implies$ products of X with other regular spaces are also regular

normal if every pair of disjoint closed sets is separated by neighborhoods

- $\implies$ every metrizable space is normal
- $\implies$ every compact T_2 space is normal

T_2 if every pair of disjoint points is separated by neighborhoods

- $\implies$ compact subsets of X are closed
- $\implies$ limits of sequences of points in X are unique
- $\implies$ subspaces of X are also T_2
- $\implies$ products of X with other T_2 spaces are also T_2

Special topologies

trivial $T = \{\varnothing, X\}$

discrete $T = \mathcal{P}(X)$

metric for metric space (X, d), the topology generated by $\{B_d(p, r) \mid p \in X \wedge r \in \mathbb{R}\}$.

order for $(X, <)$, the topology generated by the collection $\mathcal{B}$ of the following subsets $\forall a, b \in X$.

- (a, b)
- $[\min(X), b)$
- $(a, \max(X)]$

product for a collection of topological spaces of the form (X, T_X), the collection of their underlying sets $\mathcal{X}$, the topology generated by $\left\{\prod_{i\in I} U_i \mid U \in T_i\right\}$.

on $\mathbb{R}, \forall x, y \in \mathbb{R}$:

standard the metric topology induced by $d(x, y) = \sqrt{x^2 + y^2}$

lower generated by the basis of $[a, b)$

upper generated by the basis of $(a, b]$

k-topology generated by the basis of $(a, b) \setminus \left\{\frac{1}{z} \mid z \in \mathbb{Z}_+\right\}$

Maps

injective for $f : A \to B$ if $\forall a, b \in A, f(a) = f(b) \implies a = b$

surjective if $\forall b \in B, \exists a \in A : f(b) = a$

bijective if both injective and surjective

continuous a map $f : X \to Y$ between topological spaces (X, T) and (Y, S) if $\forall V \in S, f^{-1}(V) \in T$. Continuous maps preserve

- connectedness
- path-connectedness
- compactness

homeomorphism a bijection $f : X \to Y$ if both f and f^{-1} are continuous

closed map $f : X \to Y$ if $\forall A \subset X, A$ closed $\implies f(A)$ closed

Metric spaces

metric on a set X, the function $d : X \times X \to \mathbb{R}$ such that, $\forall x, y, z \in X$,

- $d(x, y) \geq 0$
- $d(x, y) = 0 \iff x = y$
- $d(x, y) = d(y, x)$
- $d(x, z) \leq d(x, y) + d(y, z)$

open ball $B_d(p, r) := \{x \mid d(x, p) < r\}$

A metric *induces* the topology generated by the basis of all open balls using d.

For $A \subset X$,

- $d(x, A) := \inf \{d(x, a) \mid a \in A\}$
- $\operatorname{diam} A := \sup \{d(a_1, a_2) \mid a_1, a_2 \in A\}$

uniformly continuous a map $f : X \to Y$ between metric spaces if $\forall \varepsilon > 0, \exists \delta > 0 : \forall x_0, x_1 \in X, d_X(x_0, x_1) < \delta \implies d_Y(f(x_0), f(x_1)) < \varepsilon$

ε-neighborhood the set

$$\begin{aligned} U(A, \varepsilon) &:= \{x \mid d(x, A) < \varepsilon\} \\ &= \bigcup_{a \in A} B_d(a, \varepsilon) \end{aligned}$$

square metric $d(p, q) = \max_i |p_i - q_i|$

Theorems

separation/subspace If X is separated by C and D and Y is a connected subspace of X, then Y lies entirely within either C or D.

closure $\operatorname{Cl}(Y) = Y \cup Y'$

closed/limit points Y is closed $\iff Y \subseteq Y'$

finite product connectedness A finite product of connected spaces is connected

Uniform limit For a sequence of functions (f_n) from a topological space to a metric space, uniform convergence to $f \implies f$ is continuous.

Intermediate value For continuous map $f : X \to Y$ from connected space X to ordered set Y in the order topology, $a, b \in X \wedge r \in Y \wedge f(a) < r < f(b) \implies \exists c \in X : f(c) = r$

Extreme value For continuous map $f : X \to Y$ where X is compact and Y is an ordered set in the order topology, then $\forall x \in X, \exists c, d \in X : f(c) \leq f(x) \leq f(d)$

Lebesgue covering For a compact metric space X with open cover $\mathcal{C}$, there is some number $\delta > 0$ such that every subset of X with diameter less than δ is contained in some member of $\mathcal{C}$.

Uniform continuity $f : X \to Y$ is a continuous map from the compact metric space (X, d_X) to the metric space $(Y, d_Y) \implies f$ is uniformly continuous.

Pf. $\mathcal{B} := \{B_{d_Y}(y, \varepsilon/2) \mid y \in Y\}$ for some $\varepsilon > 0$ is an open covering of Y. Then $\mathcal{A} := \left\{f^{-1}(B) \mid B \in \mathcal{B}\right\}$ is an open covering of X, with Lebesgue number δ. Then

$$\begin{aligned} &(x_1, x_2 \in X) \wedge (d_X(x_1, x_2) < \delta) \\ &\quad \implies \operatorname{diam} \{x_1, x_2\} < \delta \\ &\quad \implies \{f(x_1), f(x_2)\} \in B \in \mathcal{B} \\ &\quad \implies d_Y(f(x_1), f(x_2)) < \varepsilon \end{aligned}$$

l.u.b./compactness For totally ordered set X with the least upper bound property, closed intervals are compact in the order topology.

compact T_2/uncountable A nonempty compact T_2 space without isolated points is uncountable.

Miscellanea

total order a relation $<$ on a set S such that $\forall a, b, c \in S$:

- $a < a$
- $a < b \wedge b < c \implies a < c$
- $a < b \wedge b < a \implies a = b$

linear continuum a totally ordered set L such that:

- every subset bounded above has a least upper bound (l.u.b. property)
- $\forall a, b \in L, a < b \implies \exists c \in L : a < c < b$ (density)

path between points p and q in X, a continuous map $\phi : [0, 1] \to X$ such that $\phi(0) = p$ and $\phi(1) = q$